Math 526, S12
Exam 1

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	24	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3})$.

(a) Find the length of $\mathbf{v}$ and $\mathbf{w}$.

$$\|V\| = \sqrt{V\cdot V} = \sqrt{3^2+(\sqrt{3})^2+(2\sqrt{6})^2} = \sqrt{36} = 6$$

$$\|W\| = \sqrt{W\cdot W} = \sqrt{0^2+(-\frac{\sqrt{3}}{3})^2+(\frac{\sqrt{6}}{3})^2} = \sqrt{1} = 1$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\cos\theta = \frac{V\cdot W}{\|V\|\,\|W\|} = \frac{3\cdot 0+\sqrt{3}\cdot(-\frac{\sqrt{3}}{3})+2\sqrt{6}\cdot\frac{\sqrt{6}}{3}}{6\cdot 1}$$

$$= \frac{0+(-1)+4}{6} = \frac{3}{6} = \frac{1}{2}$$

$$\Rightarrow \theta = \arccos(\tfrac{1}{2}) = \frac{\pi}{3}$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$u = \frac{V}{\|V\|} = \frac{(3, \sqrt{3}, 2\sqrt{6})}{6} = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{6}, \tfrac{\sqrt{6}}{3})$$

(d) Find a number α so that $\mathbf{w} + \alpha\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$(W+\alpha V)\cdot V = 0$$

$$W\cdot V + \alpha(V\cdot V) = 0$$

$$\alpha = -\frac{W\cdot V}{V\cdot V} = -\frac{3}{36} = -\frac{1}{12}$$

Problem 2. (20 points)

(a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+2y \\ 2x-3y \end{bmatrix}$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$$

(b) Find an 3×3 elimination matrix $\mathbf{E}$ that adds 3 times row 2 to row 3.

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

(c) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 2 and 3, then rows 1 and 2.

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(d) Find two 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{0}$ but $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ -2 & 0 \end{bmatrix}$$

(There are many different answers)

(e) Find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it is invertible.

$$\frac{1}{1\cdot 4 - 3\cdot 2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Problem 3. (24points)

(a) Find the **LU** factorization of $\mathbf{A} = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$

$$A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=-2} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{31}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$$

$$\Rightarrow A = LU = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$$

(b) Given $\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find $\mathbf{A}^{-1}$ using the Gaussian-Jordan Elimination.

$$[A \ I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\ell_{21}=1} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{\ell_{31}=2} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array}\right] \xrightarrow{\ell_{13}} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{\ell_{12}=} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array}\right] \xrightarrow{\text{scale}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \end{array}\right] = [I \ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Problem 4. (18 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$.

(a) Reduce $\mathbf{A}$ to the *reduced echelon form* $\mathbf{R}$,

$$A=\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 3 & 6 & 7 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -2 & -2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

(b) Find the rank of $\mathbf{A}$.

$r = \text{rank}(A) = 2$

(c) Find the null space of $\mathbf{A}$, $N(\mathbf{A})$.

Pivot variables : x_1, x_3, free variables : x_2, x_4.

$Ax = 0 \Leftrightarrow Rx = 0$

$$\begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

i) Let $x_2 = 1$, $x_4 = 0$, we get $x_1 = -2$, $x_3 = 0$

$S_1 = (-2, 1, 0, 0)$

ii) Let $x_2 = 0$, $x_4 = 1$, we get $x_1 = -1$, $x_3 = -1$

$S_2 = (-1, 0, -1, 1)$

Thus $N(A) = \left\{ c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} -2c_1 - c_2 \\ c_1 \\ -c_2 \\ c_2 \end{bmatrix} \right\}$

Problem 5. (18 points) True or False (no explaination needed).

(a) If a matrix $\mathbf{A}$ is invertible, then $\mathbf{A}^3$ must be invertible.

True

$$\left[\begin{aligned} (A^3)^{-1} &= (A \cdot A \cdot A)^{-1} \\ &= A^{-1} \cdot A^{-1} \cdot A^{-1} \\ &= (A^{-1})^3 \end{aligned} \right]$$

(b) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, there is an 3×3 matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.

False.

$$\left[A \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ 0 & 0 & 0 \\ 2g & 2h & 2i \end{bmatrix} \neq I \right]$$

(c) The set of all vectors (b_1, b_2, b_3) with $b_1 \leq b_2 \leq b_3$ is a subspace of $\mathbb{R}^3$.

False.

[$(1, 2, 3)$ is an element but $(-1) \cdot (1, 2, 3) = (-1, -2, -3)$ is not]

(d) If a subspace of the vector space $\mathbf{M}$ contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix $\mathbf{I}$.

True

$$\left[\frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \right]$$

(e) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric, then their product $\mathbf{AB}$ is symmetric.

False

[$(AB)^T = B^T A^T = BA \neq AB$ in general

for example:

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$$]

(f) If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in $\mathbb{R}^n$, then the matrix $\mathbf{uv}^T$ is a rank one matrix.

False

[If $v = 0$, then uv^T is a zero matrix not rank one.]